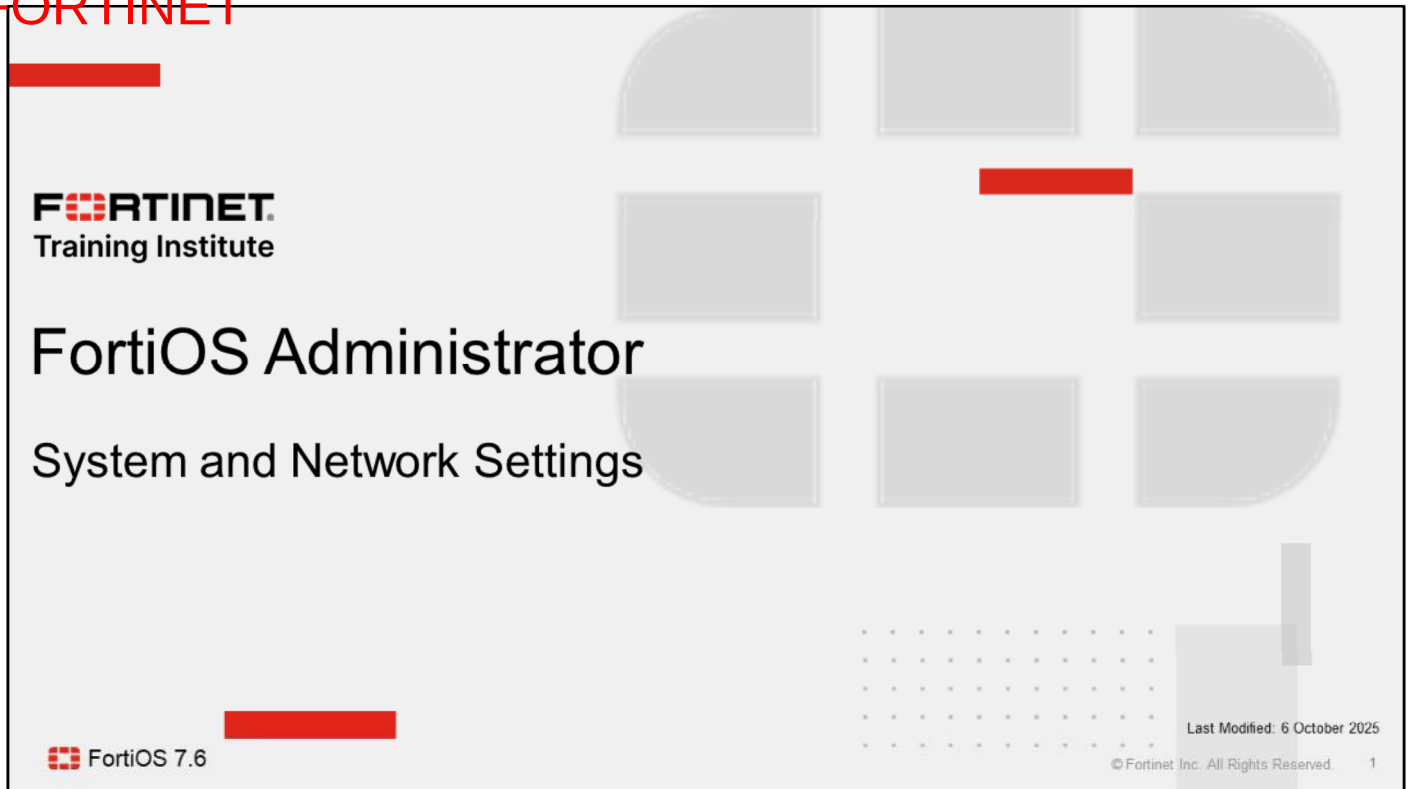


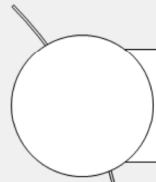
DO NOT REPRINT
© FORTINET



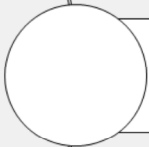
In this lesson, you will learn about system and network settings on FortiGate.

DO NOT REPRINT
© FORTINET

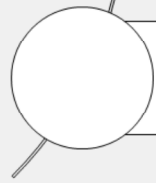
Lesson Overview



Initial Setup



Basic Administration



FortiGuard Subscription



© Fortinet Inc. All Rights Reserved. 2

In this lesson, you will learn about the topics shown on this slide.

DO NOT REPRINT
© FORTINET

Initial Setup

Objectives

- Configure FortiGate using the factory default settings
- Configure FortiGate as the DHCP server



© Fortinet Inc. All Rights Reserved. 3

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in basic system and network administration, you will be able to perform the initial setup of FortiGate and configure basic networking settings.

DO NOT REPRINT
© FORTINET

The Modern Context of Network Security

- Firewalls are more than gatekeepers on the network perimeter
- Today's firewalls are designed in response to multifaceted and multidevice environments with no identifiable perimeter:
 - Mobile workforce
 - Partners accessing your network services
 - Public and private clouds
 - Internet of Things (IoT)
 - Operation technology (OT)
 - Bring your own device (BYOD)
- Firewalls are expected to perform different functions within a network
 - Different deployment modes:
 - Distributed enterprise firewall
 - Next-generation firewall (NGFW)
 - Internal segmentation firewall (ISFW)
 - Data-center firewall
 - DNS, DHCP, web filter, intrusion prevention system (IPS), and so on



© Fortinet Inc. All Rights Reserved.

4

In the past, the most common method of protecting a network was securing the perimeter and installing a firewall at the entry point. Network administrators trusted everything and everyone inside the perimeter.

Now, malware can easily bypass any entry-point firewall and get inside the network. This could happen through an infected USB stick or a compromised device being connected to the corporate network. Additionally, because attacks can come from inside a network, network administrators can no longer inherently trust internal all users and devices.

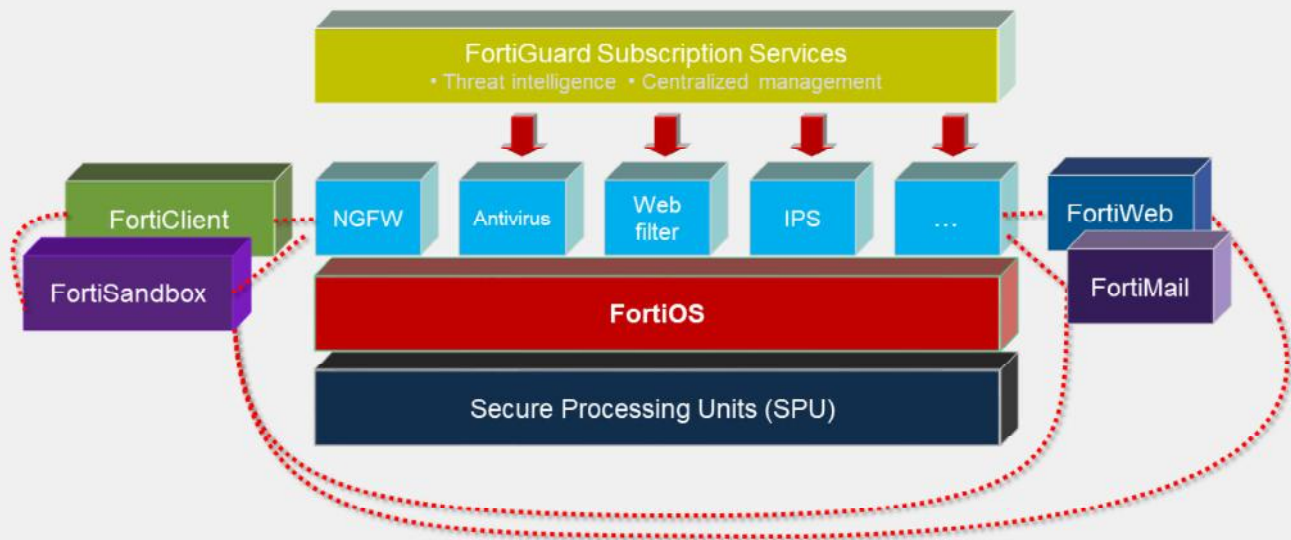
What's more, today's networks are highly complex environments whose borders are constantly changing. Networks run vertically from the LAN to the internet, and horizontally from the physical network to the private virtual network and to the cloud. A mobile and diverse workforce (employees, partners, and customers) accessing network resources; public and private clouds; and IoT, OT, and BYOD programs all conspire to increase the number of attack vectors faced by your network.

In response to this highly complex environment, firewalls have become robust, multifunctional devices that counter an array of threats to your network. FortiGate can act in different modes or roles to address different requirements. For example, FortiGate can be deployed as a data-center firewall whose function is to monitor inbound requests to servers and to protect them without increasing latency for the requester. Alternatively, FortiGate can be deployed as an ISFW as a way to contain a network breach.

FortiGate can also function as DNS and DHCP servers and be configured to provide web filter, antivirus, and IPS services.

DO NOT REPRINT
© FORTINET

Platform Design



© Fortinet Inc. All Rights Reserved. 5

In the architecture diagram shown on this slide, you can see how FortiGate platforms add strength, without compromising flexibility. Like separate, dedicated security devices, FortiGate is still *internally* modular. Plus:

- Devices add duplication. Sometimes, dedication *doesn't* mean efficiency. If it's overloaded, can one device borrow free RAM from nine others? Do you want to configure policies, logging, and routing on 10 separate devices? Does 10 times the duplication bring you 10 times the benefit, or is it a hassle? For small and midsize businesses (SMB) or enterprise branch offices, unified threat management (UTM) is often a superior solution compared to separate, dedicated appliances.
- FortiGate hardware isn't just off-the-shelf. It's carrier grade. Most FortiGate models have one or more specialized circuits, called application-specific integrated circuits (ASIC), that are engineered by Fortinet. For example, a content processor (CP) or network processor (NP) chip handles cryptography and packet forwarding more efficiently. Compared to a single-purpose device with only a CPU, FortiGate can have dramatically better performance. This is especially critical for data centers and carriers where throughput is business critical.
(The exception? Virtualization platforms—VMware, Citrix Xen, Microsoft, or Oracle Virtual Box—have general-purpose virtual CPUs (vCPUs). But, virtualization might be worthwhile because of other benefits, such as distributed computing and cloud-based security.)
- FortiGate is flexible. If all you need is fast firewalling and antivirus, FortiGate will not cause you to waste CPU, RAM, and electricity on other features. In each firewall policy, you can enable or disable UTM and NGFW modules. Also, you won't pay more to add VPN seat licenses later.
- FortiGate cooperates. A preference for open standards instead of proprietary protocols means less vendor lock-in and more choice for system integrators. And, as your network grows, FortiGate can leverage other Fortinet products, such as FortiSandbox and FortiWeb, to distribute processing for deeper security and optimal performance—a total Security Fabric approach.

DO NOT REPRINT
© FORTINET

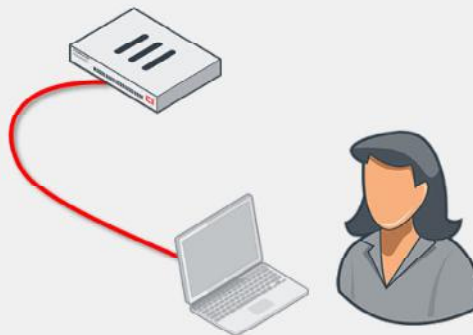
Factory Default Settings

- IP: 192.168.1.99/24
 - Management interface on high-end and mid-range models
 - Port1 or internal interface on entry-level models
- PING, HTTPS, and SSH protocol management enabled
- Built-in DHCP server is enabled on port1 or internal interface
 - Only on entry-level models that support DHCP server
- Default login

User: admin

Password: (blank)

 - Both are case sensitive
 - Modify the default (blank) password
- Can access FortiGate on the CLI
 - Console: without network
 - CLI console widget and terminal emulator, such as PuTTY or Tera Term



© Fortinet Inc. All Rights Reserved. 5

Network address translation (NAT) mode is the default operation mode. What are the other factory default settings? After you have removed FortiGate from its box, what do you do next?

Attach your computer network cable to port1 or the internal switch ports (on the entry-level model). For high-end and mid-range models, connect to the management interface. In most entry-level models, there is a DHCP server on that interface. So, if your computer's network settings have DHCP enabled, your computer should automatically get an IP address, and you can begin to set up.

To access the GUI on FortiGate or FortiWiFi, open a web browser and visit <https://192.168.1.99>.

The default login information is public knowledge. Never leave the default password blank. Your network is only as secure as your FortiGate admin account. Once you log in with default login details, you'll see a message instructing you to the default admin user password. Before you connect FortiGate to your network, you should set a complex password. You will also be asked to apply additional configuration information such as host name, dashboard setup, FortiCare registration, and so on.

All FortiGate models have a console port, or USB management port, or both. The port provides CLI access, without a network. You can access the CLI using the CLI console widget on the GUI or from a terminal emulator, such as PuTTY or Tera Term.

Modes of Operation

NAT mode (default operation mode)

- FortiGate is an OSI layer 3 *router*
- Interfaces have IP addresses
- Packets are routed by IP address



Transparent mode

- FortiGate is an OSI layer 2 *switch or bridge*
- Interfaces do *not* have IP addresses
- Cannot route packets, only forward or block



When you deploy FortiGate, you can choose between two operating modes: NAT mode or transparent mode.

- In NAT mode, FortiGate routes packets based on layer 3, like a router. Each of its logical network interfaces has an IP address, and FortiGate determines the outgoing or egress interface based on the destination IP address and entries in its routing tables.
- In transparent mode, FortiGate forwards packets at layer 2, like a switch. Its interfaces have no IP addresses, and FortiGate identifies the outgoing or egress interface based on the destination MAC address. The device in transparent mode has an IP address used for management traffic.

NAT mode is the default operation mode. In this course, you will learn about only the NAT mode operation of FortiGate.

DO NOT REPRINT
© FORTINET

Interface IP Addresses

- In NAT mode, you can't use interfaces until they have an IP address:
 - Manually assigned
 - Automatic
 - DHCP
 - PPPoE

Network > Interfaces

Edit Interface

Name

Alias

Type ☒ Physical Interface

VRF ID

Role

☐ Dedicated Management Port

Address

Addressing mode ☒ Manual ☐ IPAM ☐ DHCP ☐ PPPoE ☐ One-Arm Sniffer

IP/Netmask

Secondary IP address ☐

Network > Interfaces

Edit Interface

Name

Alias

Type ☒ Physical Interface

VRF ID

Role

☐ Dedicated Management Port

Address

Addressing mode ☐ Manual ☐ IPAM ☒ DHCP ☐ PPPoE ☐ One-Arm Sniffer

Retrieve default gateway from server ☒

Distance

Override internal DNS ☒



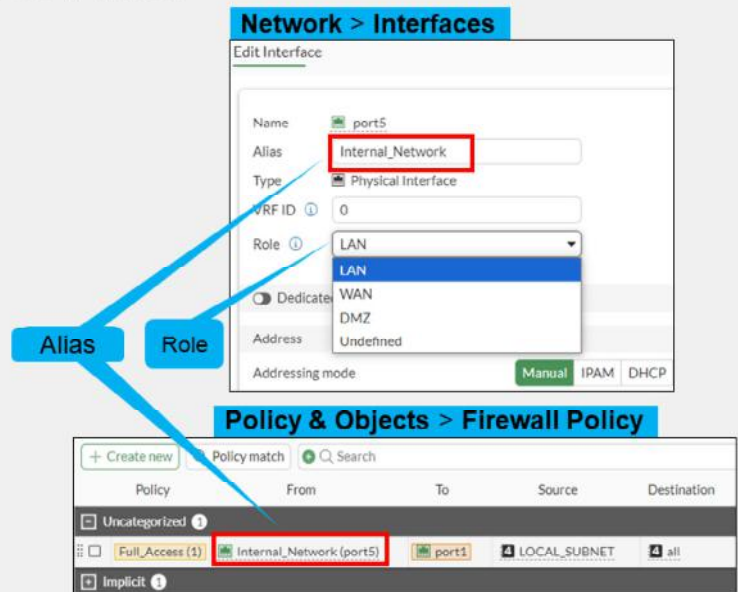
When FortiGate is operating in NAT mode, every interface that handles traffic must have an IP address. When in NAT mode, FortiGate can use the IP address to source the traffic, if it needs to start or reply to a session, and as a destination address for devices trying to contact FortiGate or route traffic through it. There are multiple ways to get an IP address:

- Manually
- Automatically, using either DHCP or Point-to-Point Protocol over Ethernet (PPPoE) (available on the CLI)

DO NOT REPRINT
© FORTINET

Interface Role Compared to Alias

- Role defines interface settings typically grouped together:
 - Prevents accidental misconfiguration
 - Four types:
 - LAN
 - WAN
 - DMZ
 - Undefined (show all settings)
 - Not in a list of policies
- Alias is a friendly descriptor for the interface:
 - Used in a list of policies to label interfaces by purpose



How many times have you seen network issues caused by a DHCP server—not client—enabled on the WAN interface?

You can configure the interface role. The roles shown on the GUI are the usual interface settings for that part of a topology. Settings that do not apply to the current role are hidden on the GUI. (All settings are always available on the CLI, regardless of the role.) This prevents accidental misconfiguration.

For example, when the role is configured as **WAN**, there is no DHCP server and device detection configuration available. Device detection is usually used to detect devices internally on your LAN.

If there is an unusual case, and you need to use an option that's hidden by the current role, you can always switch the role to **Undefined**. This displays all options.

To help you remember the purpose of each interface, you can give them aliases. For example, you could call `port3 internal_network`. This can make your list of policies easier to comprehend.

Administrative access options are also limited depending on the role that is set for the interface. For an example, setting the interface role to **WAN** would not display the **Ping** option mitigating the risk of responding to a DoS ICMP attack from the WAN.

DO NOT REPRINT
© FORTINET

FortiGate as a DHCP Server

Network > Interfaces

Edit Interface

Name: port3
Alias: LAN
Type: Physical Interface
VRF ID: 0
Role: LAN
Addressing mode: **Manual** IPAM DHCP
IP/Netmask: 10.0.1.254/255.255.255.0
Create address object matching subnet: ☒
Name: port3 address
Destination: 10.0.1.0/24
Secondary IP address: ☐

Administrative Access

IPv4: ☒ HTTPS ☒ HTTP ☒ PING
☐ FMG-Access ☐ SSH ☐ SNMP
☐ FTM ☐ RADIUS Accounting ☐ Security Fabric Connection
☐ Speed Test
Receive LLDP: Use VDOM Setting **Enable** Disable
Transmit LLDP: Use VDOM Setting **Enable** Disable

DHCP Server

DHCP status: **Enabled** Disabled
Address range: 10.0.1.1-10.0.1.253
Netmask: 255.255.255.0
Default gateway: Same as Interface IP Specify
DNS server: Same as System DNS Same as Interface IP Specify
Lease time: 604800 second(s)
Advanced

For an interface (such as port3), select **Manual**, enter a static IP address, and then enable **DHCP Server**. Options for the built-in DHCP server appear, including provisioning features, such as DHCP options and IP address assignment rules.

Advanced DHCP options allow you to automatically configure clients with additional network settings, such as DNS servers or NTP configurations. You can also enable DHCP relay to forward requests to an external server which is useful when managing IP addresses across multiple subnets. Additionally, custom IP address assignment rules provide precise control by mapping specific IP addresses to MAC addresses, or allocating address ranges, preventing conflicts, and ensuring consistency across the network.

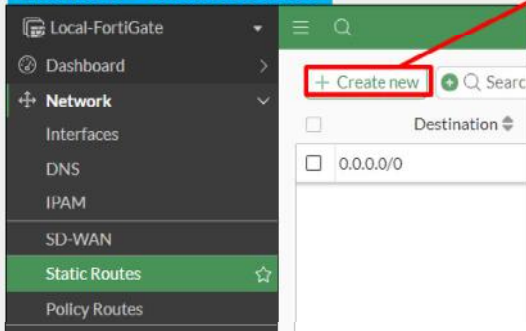
DO NOT REPRINT

© FORTINET

Static Gateway

- Set a default route or configure a static one
- If the interface is DHCP or PPPoE, the gateway can be set dynamically

Network > Static Routes



New Static Route

Destination ? Subnet Internet Service

Gateway Address

Interface port1 + ×

Administrative Distance ?

Comments

Status Enabled Disabled

Advanced Options

Priority ?



Before you integrate FortiGate into your network, you should configure a default gateway.

If FortiGate gets its IP address through a dynamic method, such as DHCP or PPPoE, then it should also retrieve the default gateway.

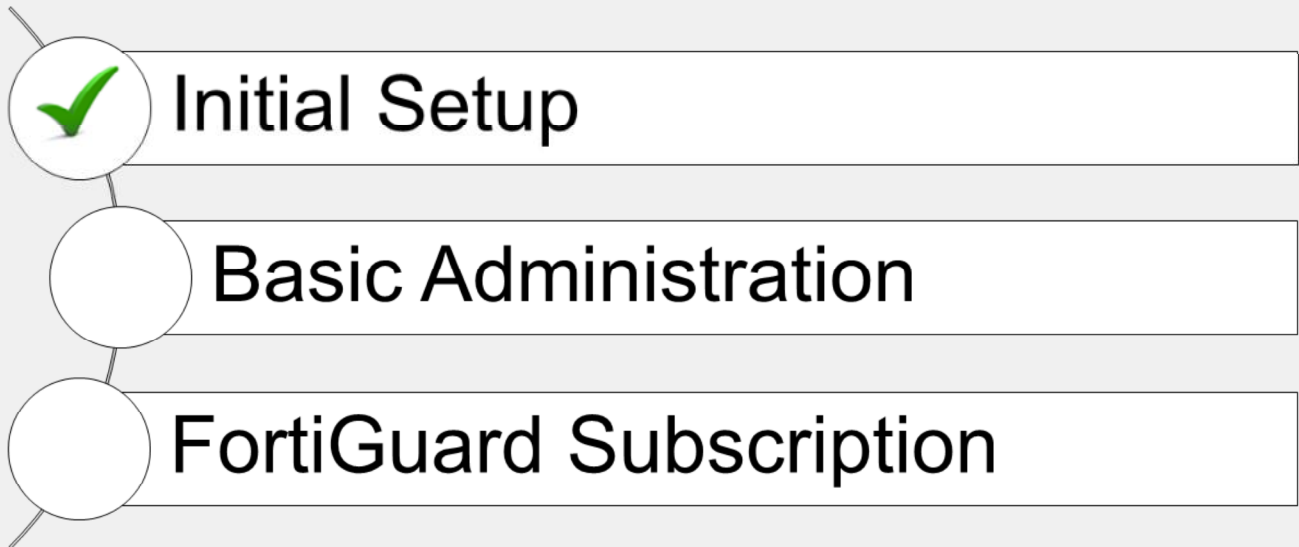
Otherwise, you must configure a static route. Without this, FortiGate will not be able to respond to packets outside the subnets directly attached to its own interfaces. It likely will not be able to connect to FortiGuard—which is crucial for FortiGate updates—and may fail to route traffic correctly.

Make sure that FortiGate has a default route (matching all packets with the destination 0.0.0.0/0) configured to forward traffic through the network interface connected to the internet, directing it to the IP address of the next router.

You must configure routing to establish essential network settings before you can configure firewall policies.

DO NOT REPRINT
© FORTINET

Lesson Progress



Good job! You now understand how to perform the initial setup of FortiGate.

Now, you will learn about basic administration.

DO NOT REPRINT
© FORTINET

Basic Administration

Objectives

- Configure and control administrator access to FortiGate

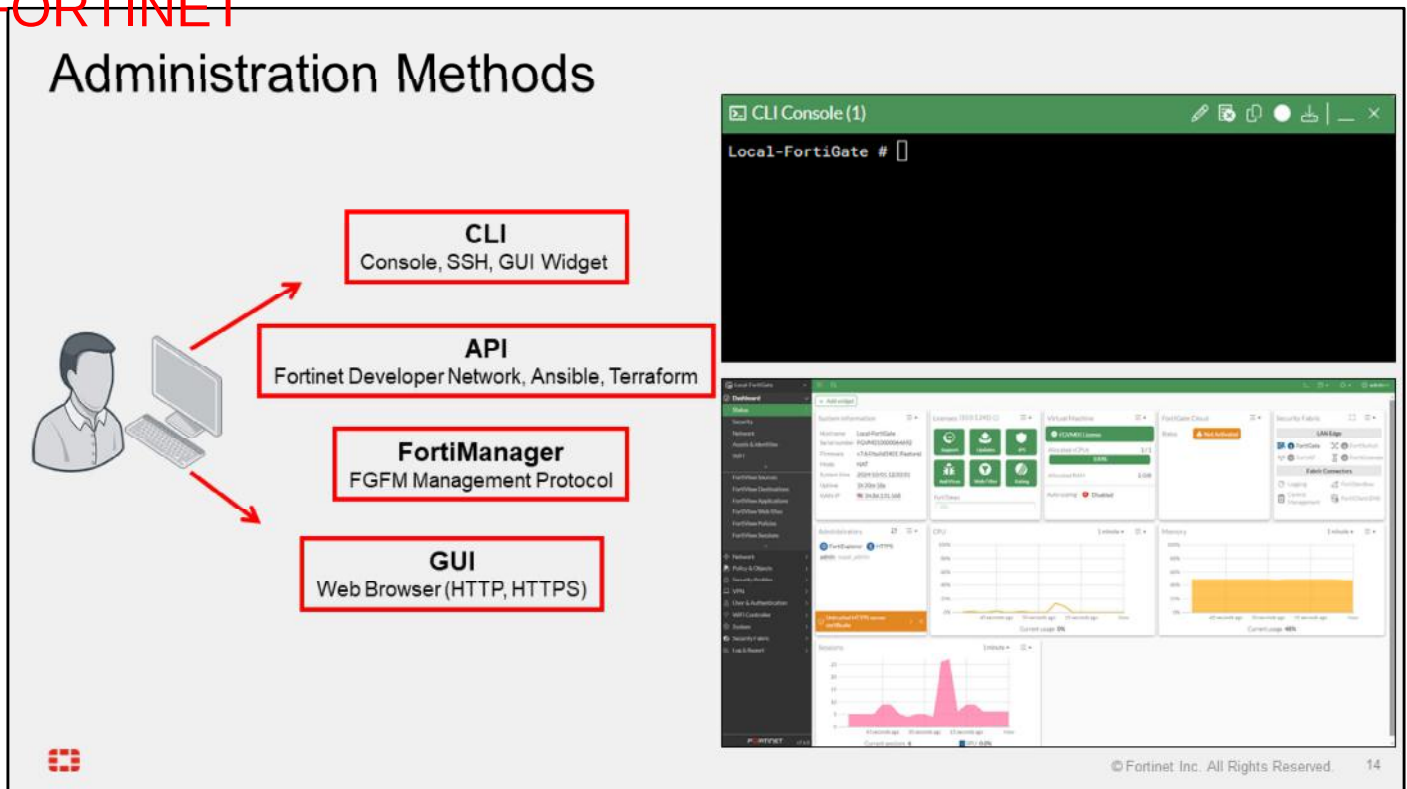


© Fortinet Inc. All Rights Reserved. 13

After completing this section, you should be able to achieve the objective shown on this slide.

By demonstrating competence in basic administration, you will be able to better manage administrative users and implement stronger security practices for administrative access.

DO NOT REPRINT
© FORTINET



Most features are available on both the GUI and CLI, but there are a few exceptions. You can't view reports on the CLI. Also, advanced settings and diagnostic commands for super users are usually not available on the GUI.

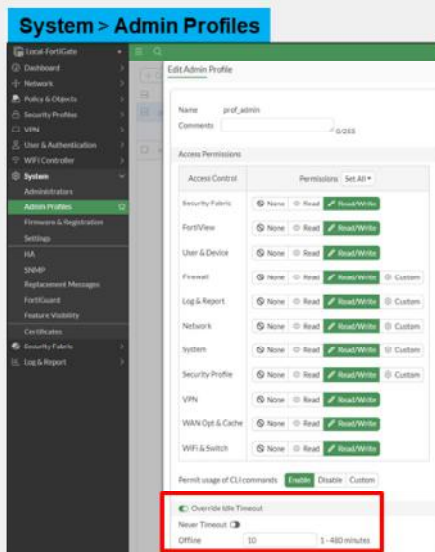
As you become more familiar with FortiGate, and especially if you want to script its configuration, you might want to use the CLI, API or FortiManager in addition to the GUI. You can access the CLI through either the JavaScript widget on the GUI named **CLI Console**, or through a terminal emulator such as Tera Term or PuTTY. Your terminal emulator can connect through the network—SSH—or the local console port.

SNMP and some other administrative protocols are also supported, but they are read-only. You can't use them for basic setup.

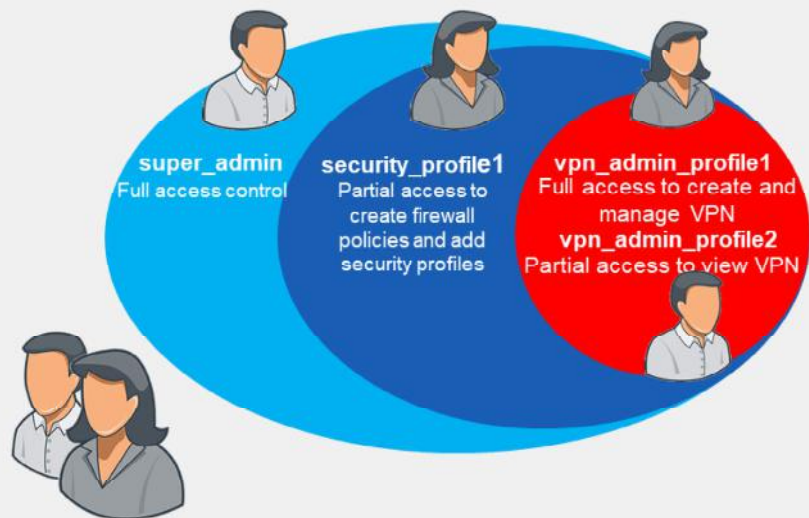
DO NOT REPRINT
© FORTINET

Administrator Profiles

• Permissions



• Hierarchy



When assigning permissions to an administrator profile, you can specify read-and-write, read-only, or none to each area.

By default, there is a special profile named `super_admin`, which is used by the account named `admin`. You can't change it. It provides full access to everything, making the `admin` account similar to a root superuser account.

You aren't required to use a default profile. You could create a profile named `auditor_access` with read-only permissions. Restricting a person's permissions to those necessary for his or her job is a best practice, because even if that account is compromised, the compromise to your FortiGate device (or network) is not total. To do this, create administrator profiles and then select the appropriate profile when configuring an account.

The **Override Idle Timeout** option allows the `admintimeout` value, under `config system accprofile`, to be overridden for each access profile. You can configure administrator profiles to increase inactivity timeout and facilitate use of the GUI for central monitoring. Note that you can do this on a per-profile basis, to prevent the option from being unintentionally set globally. So, what are the effects of administrator profiles?

It's actually more than just read or write access. Depending on the type of administrator profile that you assign, an administrator may not be able to access the entire FortiGate device. For example, you could configure an account that can view only vpn settings and configure another profile to create firewall policies and security profiles.

DO NOT REPRINT
© FORTINET

Administrative Access—Trusted Sources

System > Administrators

☐ Two-factor Authentication

☒ Restrict login to trusted hosts

Trusted Host 1

☐ Restrict admin to guest account provisioning only

+ Create new

Search

☐	Na...	Trusted Hosts	Profile	Type	Two-factor Authentication
☒	System Administrator				
☐	admin	10.0.1.10/32	super_admin	Local	Disabled

You can restrict an admin user to manage guest users with a guest group in place to provision users

If the admin user attempts to log in to the FortiGate GUI from any IP address other than 10.0.1.10, they receive this message

Authentication failure

Username

Password



© Fortinet Inc. All Rights Reserved. 16

Another way to secure FortiGate is to define the hosts or subnets that are trusted sources from which to log in.

In this example, 10.0.1.10 is configured as the only trusted IP address for sources that the administrator logs in from. If the administrator attempts to log in from a machine with any other IP address, they will receive an authentication failure message.

Note that if trusted hosts are configured on all administrators and an administrator is trying to log in from an IP address that is not set on any of the trusted hosts for any administrators, then the administrator will not get the login page.

If you leave any IPv4 address as 0.0.0.0/0, it means that connections from any source IP address will be allowed. By default, 0.0.0.0/0 is the configuration for the administrator, although you may want to change this.

Note that each account can define its management host or subnet differently. Be aware of any NAT that occurs between the specified device and FortiGate. You can easily prevent an administrator from logging in from the specified IP address if it is later NATed to another address before reaching FortiGate, thus defeating the purpose of the trusted hosts.

You can also configure an administrator account that is restricted to provisioning guest accounts only. This option requires you to configure a guest user group.

DO NOT REPRINT
© FORTINET

Administrative Access—Ports and Password

- Port numbers are customizable
- Fortinet recommends using only secure access (SSH, HTTPS)
- Default **Idle timeout** value is 5 minutes

System > Settings

The screenshot shows the 'System > Settings' page in FortiOS. The 'Administration Settings' section is highlighted with a red box and contains the following fields:

Setting	Value
HTTP port	80
Redirect to HTTPS	☐
HTTPS port	443
HTTPS server certificate	self-sign
SSH port	22
Telnet port	23
Idle timeout	5 Minutes (1 - 480)
ACME interface	+
Allow concurrent sessions	☒
FortiCloud Single Sign-On	☐

The 'Password Policy' section is also highlighted with a red box and contains the following fields:

Setting	Value
Password scope	Off Admin IPsec Both
Minimum length	8
Minimum number of new characters	0
Character requirements	☐
Allow password reuse	☒
Password expiration	☐



© Fortinet Inc. All Rights Reserved. 17

You may also want to customize the administrative protocols port numbers.

You can choose whether to allow concurrent sessions. You can disallow concurrent sessions to avoid accidentally overwriting settings. Allowing concurrent settings lets an administrator open multiple browser tabs and CLI sessions at the same time. This may be more convenient for some situations, but it may also result in an administrator accidentally editing the same settings in conflicting ways over different connections.

For better security, use only secure protocols, and enforce password complexity and changes.

The **Idle timeout** setting specifies the number of minutes before an inactive administrator session times out (the default is 5 minutes). A shorter idle timeout is more secure, but increasing the timer can reduce the chance of administrators being logged out while testing changes.

You can override the idle timeout setting for each administrator profile using the **Override Idle Timeout** setting.

DO NOT REPRINT
© FORTINET

Administrative Access—Protocols

- Enable acceptable management protocols on each interface independently:
 - Separate IPv4 and IPv6
 - IPv6 options hidden by default
- Other protocols for which FortiGate is the destination IP address:
 - Security Fabric connection:
 - CAPWAP
 - FortiTelemetry
 - FMG-Access
 - FTM
 - RADIUS Accounting
- LLDP support
 - Detecting an upstream Security Fabric FortiGate through LLDP

Network > Interfaces

Edit Interface

Name: port4
Alias:
Type: Physical Interface
VRF ID: 0
Role: Undefined

Address

Addressing mode: Manual IPAM DHCP PPPoE
IP/Netmask: 10.0.11.254/255.255.255.0
Secondary IP address: ☐

Administrative Access

IPv4

☐ SCIM	☒ HTTPS	☒ HTTP
☒ PING	☐ FMG-Access	☒ SSH
☐ SNMP	☐ FTM	☐ RADIUS Accounting
☐ Security Fabric Connection	☐ Speed Test	

Receive LLDP: Use VDOM Setting Enable Disable
Transmit LLDP: Use VDOM Setting Enable Disable



© Fortinet Inc. All Rights Reserved. 18

You've defined the management subnet—that is, the trusted hosts—for each administrator account. How do you enable or disable management protocols?

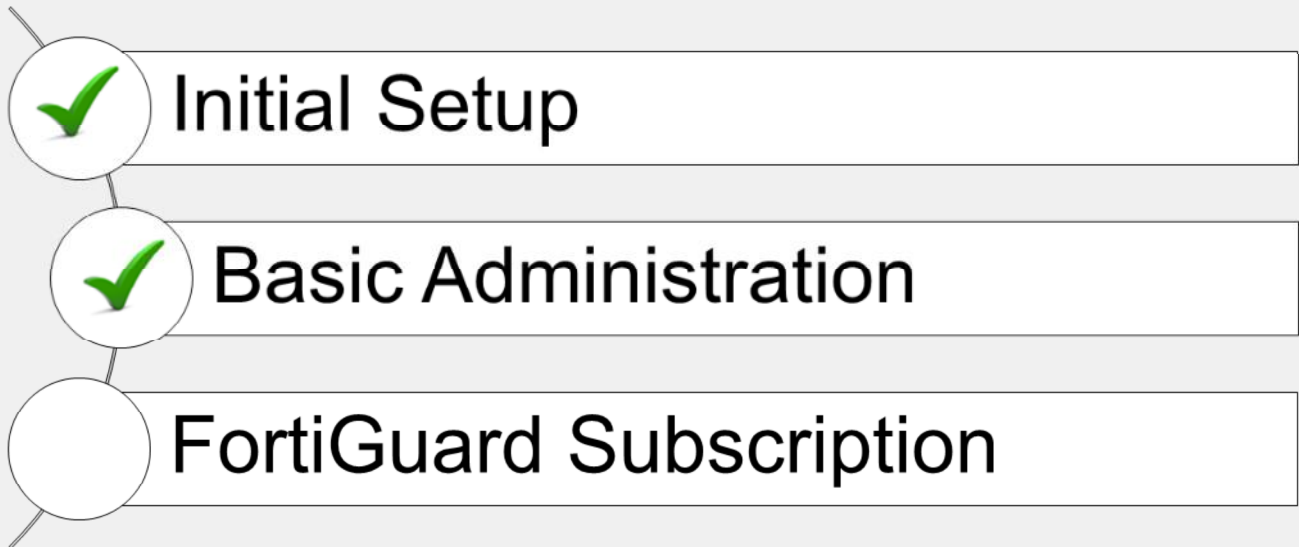
This is specific to each interface. For example, if your administrators connect to FortiGate only from port3, then you should disable administrative access on all other ports. This prevents brute force attempts and also insecure access. Your management protocols are HTTPS, HTTP, PING, and SSH. By default, the HTTP option is not visible on the GUI.

Consider the location of the interface on your network. Enabling PING on an internal interface is useful for troubleshooting. However, if it's an external interface (in other words, exposed to the internet), then the PING protocol could expose FortiGate to a denial of service (DoS) attack. You should disable protocols that do not encrypt data flow, such as HTTP. IPv4 and IPv6 protocols are separate. It's possible to have both IPv4 and IPv6 addresses on an interface, but only respond to pings on IPv6.

Security Fabric connection includes CAPWAP and FortiTelemetry. Protocols like FortiTelemetry are *not* for administrative access, but, like GUI and CLI access, they are protocols where the packets have FortiGate as a destination IP. Use the FortiTelemetry protocol specifically for managing FortiClient and the Security Fabric. Use the CAPWAP protocol for FortiAP, FortiSwitch, and FortiExtender when they are managed by FortiGate. Use the FMG-Access protocol specifically for communicating with FortiManager when that server is managing multiple FortiGate devices. Use the RADIUS accounting protocol when FortiGate needs to listen for and process RADIUS accounting packets for single sign-on (SSO) authentication. FTM, or FortiToken Mobile push, supports second-factor authentication requests from a FortiToken mobile app. When you assign the interface roles LAN or WAN to the appropriate interfaces, your FortiGate uses the Link Layer Discovery Protocol (LLDP) to detect if there's an upstream FortiGate in your network. If FortiGate discovers an upstream FortiGate, you're prompted to configure the upstream FortiGate device to join the Security Fabric.

DO NOT REPRINT
© FORTINET

Lesson Progress



Good job! You now understand basic administration.

Now, you will learn about FortiGuard subscription.

DO NOT REPRINT
© FORTINET

FortiGuard Subscription

Objectives

- Check and verify FortiGuard licenses
- Describe session packets on FortiGate



© Fortinet Inc. All Rights Reserved. 20

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in FortiGuard subscription, you will be able to ensure that FortiGate remains reliably in service throughout its life cycle.

DO NOT REPRINT
© FORTINET

FortiGuard Subscription Services

- Internet connection and contract required
- Provided by FortiGuard Distribution Network (FDN)
 - Major data centers in North America, Asia, and Europe
 - Or, from FDN through your FortiManager
 - FortiGate prefers the data center in nearest time zone, but will adjust by server load
- Package updates: FortiGuard antivirus and IPS
 - `update.fortiguard.net`
 - TCP port 443 (SSL)
- Live queries: FortiGuard web filtering, DNS filtering, and antis spam
 - `service.fortiguard.net` for proprietary protocol on UDP port 53 or 8888
 - `securewf.fortiguard.net` for HTTPS over port 443, 53 or, 8888
- FortiOS uses FortiGuard server for DNS request
 - By default, uses DNS over TLS (DoT) to secure DNS traffic



© Fortinet Inc. All Rights Reserved. 21

Some FortiGate services connect to other servers, such as FortiGuard, in order to work. FortiGuard Subscription Services provide FortiGate with up-to-date threat intelligence. FortiGate uses FortiGuard by:

- Periodically requesting packages that contain a new engine and signatures
- Querying the FDN on an individual URL or host name

By default, the FortiGuard server location is set to anywhere FortiGate selects a server based on server load, from any part of the world. However, you have the option to change the FortiGuard server location to USA. In this case, FortiGate selects a USA-based FortiGuard server.

Queries are real-time; that is, FortiGate asks the FDN every time it scans for spam or filtered websites. FortiGate queries, instead of downloading the database, because of the size and frequency of changes that occur to the database. Also, you can select queries to use UDP or HTTPs for transport; the protocols are not designed for fault tolerance, but for speed. So, queries require that your FortiGate device has a reliable internet connection.

Packages, like antivirus and intrusion prevention system (IPS), are smaller and don't change as frequently, so they are downloaded (in many cases), only once a day. They are downloaded using TCP for reliable transport. After the database is downloaded, their associated FortiGate features continue to function, even if FortiGate does not have reliable internet connectivity. However, you should still try to avoid interruptions during downloads—if your FortiGate device must try repeatedly to download updates, it can't detect new threats during that time.

When using FortiGuard servers for DNS, FortiOS uses DNS over TLS (DoT) by default to secure the DNS traffic. New FortiGuard DNS servers have been added as primary and secondary servers.

DO NOT REPRINT
© FORTINET

FortiGuard Licenses

System > FortiGuard

FortiGuard Distribution Network

License Information

Entitlement	Status	
Advanced Malware Protection	Licensed (Expiration Date: 2026/01/17)	
Attack Surface Security Rating	Licensed (Expiration Date: 2026/01/17)	
Data Loss Prevention (DLP)	Not Licensed	
Email Filtering	Licensed (Expiration Date: 2026/01/17)	
Intrusion Prevention	Licensed (Expiration Date: 2026/01/17)	
Operational Technology (OT) Security Service	Not Licensed	Purchase
Web Filtering	Licensed (Expiration Date: 2026/01/17)	
SD-WAN Network Monitor	Not Licensed	Purchase
SD-WAN Overlay as a Service	Not Licensed	Purchase
FortiSASE SPA Service Connection	Not Licensed	Purchase
FortiSASE Secure Edge Management	Not Licensed	Purchase
FortiGate Cloud	Not Activated	Activate
FortiGate Cloud Sandbox	Licensed (Expiration Date: 2026/01/17)	
FortiToken Cloud	In Trial (Expiration Date: 2024/10/08)	Upgrade
Firmware & General Updates	Licensed (Expiration Date: 2026/01/17)	
FortiCare Support	Registered	Actions
FortiConverter Service	Not Licensed	
Virtual Machine	Valid	FortiGate VM License

FortiCare support contracts can be activated here and applied directly to this FortiGate.
Enter Registration Code

FortiGuard Updates

Next Update: 2024/10/01 13:35:00

Update Licenses & Definitions Now

Manual Update

Upload License File

Fortinet Service Communications

Service	Traffic Volume (last 24 hours)
FortiCare	0 B
FortiGate Cloud Log	0 B
FortiGuard.com	556.11 kB
FortiGuard Download	8.89 kB
FortiGuard Query	7.77 kB
FortiGate Cloud Sandbox	0 B
OTIWS	0 B
FortiToken Registration	0 B
SMS Service	0 B

Additional Information

API Preview

Edit in CLI

Online Guides

Relevant Documentation

Video Tutorials

How to Purchase/Renew Fortinet Service Subscriptions

Fortinet Inc. All Rights Reserved. 22

You can check the status of FortiGuard licenses and the communication to FortiGuard on the FortiGate GUI. You can also check the versions of the locally installed databases for each of the FortiGuard services.

DO NOT REPRINT
© FORTINET

FortiGuard Licenses (Contd)

```
Local-FortiGate # diagnose autoupdate versions
AV Engine
-----
Version: 7.00030 signed
Contract Expiry Date: Mon Jan 19 2026
Last Updated using manual update on Thu Jul 13 02:54:00 2024
Last Update Attempt: Mon Aug 25 13:52:18 2024
Result: No Updates

Virus Definitions
-----
Version: 90.01635 signed
Contract Expiry Date: Mon Jan 19 2026
Last Updated using manual update on Mon Jul 25 13:52:18 2024
Last Update Attempt: Mon Aug 25 13:52:18 2024
Result: Updates Installed
```



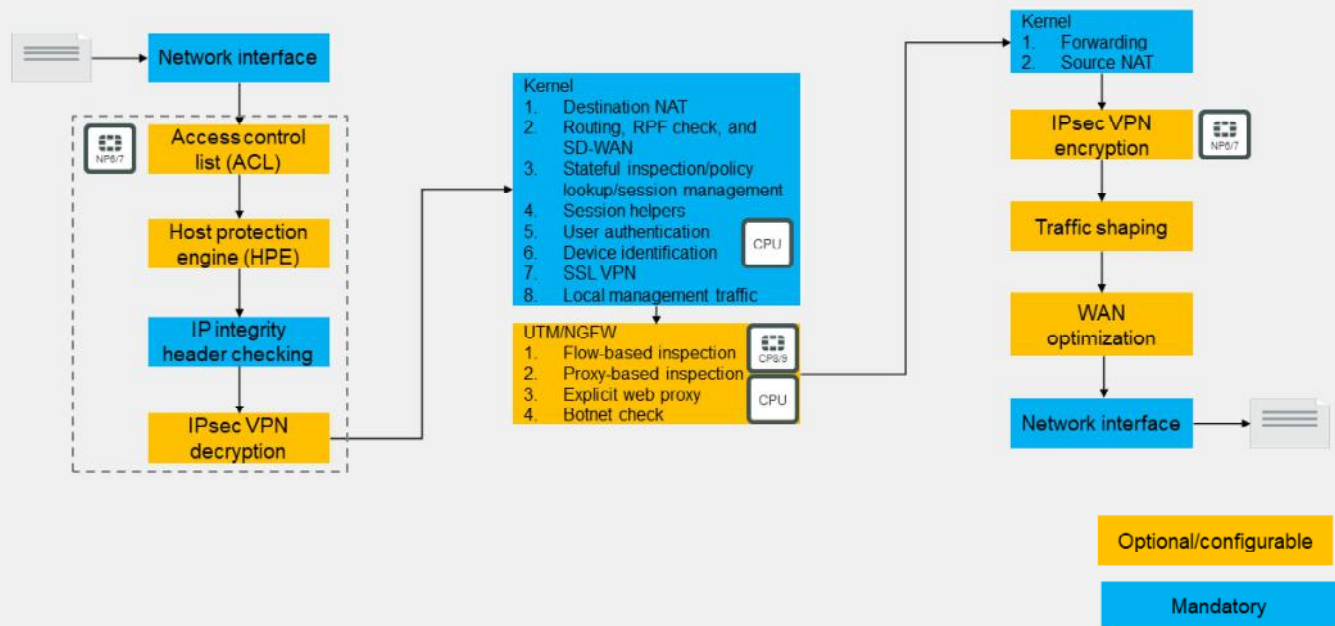
© Fortinet Inc. All Rights Reserved. 23

The command shown on this slide lists all the FortiGuard databases and engines installed. The information includes the version, contract expiration date, time it was updated, and what happened during the last update.

The list includes but is not limited to antivirus, IPS, application, mobile malware definitions, and other security services. FortiGate is licensed and updated using FortiGuard services.

DO NOT REPRINT
© FORTINET

Life of a Packet—Initial Session Packets



© Fortinet Inc. All Rights Reserved. 24

The flowchart on this slide shows the sequence of operations that the first packets of a new session go through as they enter, pass through, and exit FortiGate.

It also shows which processes can be offloaded to the network processor (NP) and content processor (CP), which are specially designed integrated circuits that improve performance and increase processing speeds on FortiGate. These processors are collectively known as secure processing units (SPU).

Many of the topics you covered in this lesson are included in this flow chart. You can use it as reference material.

DO NOT REPRINT
© FORTINET

Knowledge Check

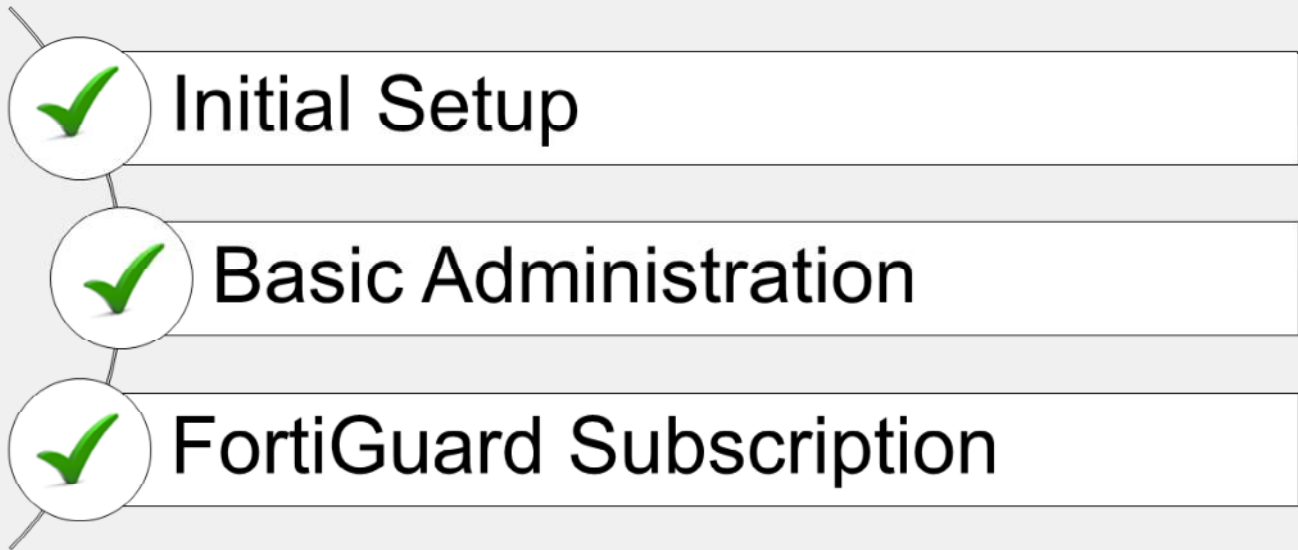
1. How do you restrict logins to FortiGate from specific IP addresses?
 - A. Change the FortiGate management interface IP address.
 - ✓ B. Configure a trusted host.

2. Which protocol option should you enable to give FortiManager access on FortiGate?
 - ✓ A. FMG-Access protocol
 - B. FTM protocol



DO NOT REPRINT
© FORTINET

Lesson Progress



Congratulations! You have completed this lesson.

Now, you will review the objectives covered in this lesson.

DO NOT REPRINT
© FORTINET

Review

- ✓ Configure FortiGate using the factory default settings
- ✓ Configure FortiGate as the DHCP server
- ✓ Configure and control administrator access to FortiGate
- ✓ Check and verify FortiGuard licenses
- ✓ Describe session packets on FortiGate



This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how and where FortiGate fits into your network and how to perform basic FortiGate administration.